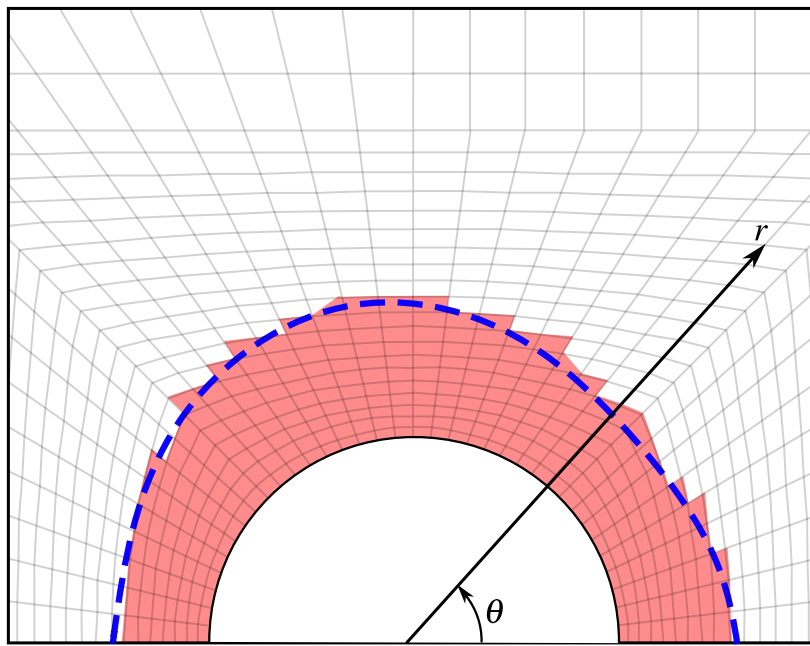


$$d_1/2R_t = 2.5, E = 20 \text{ GPa}, \nu = 0.3, c = 5 \text{ MPa}, \phi = 30^\circ$$

$$\sigma_h = 40 \text{ MPa}, \sigma_v = 30 \text{ MPa}$$



--- Limite da zona plástica - Solução analítica (Ma et al., 2020)

■ Zona plástica - Solução numérica 3D (MEF)

